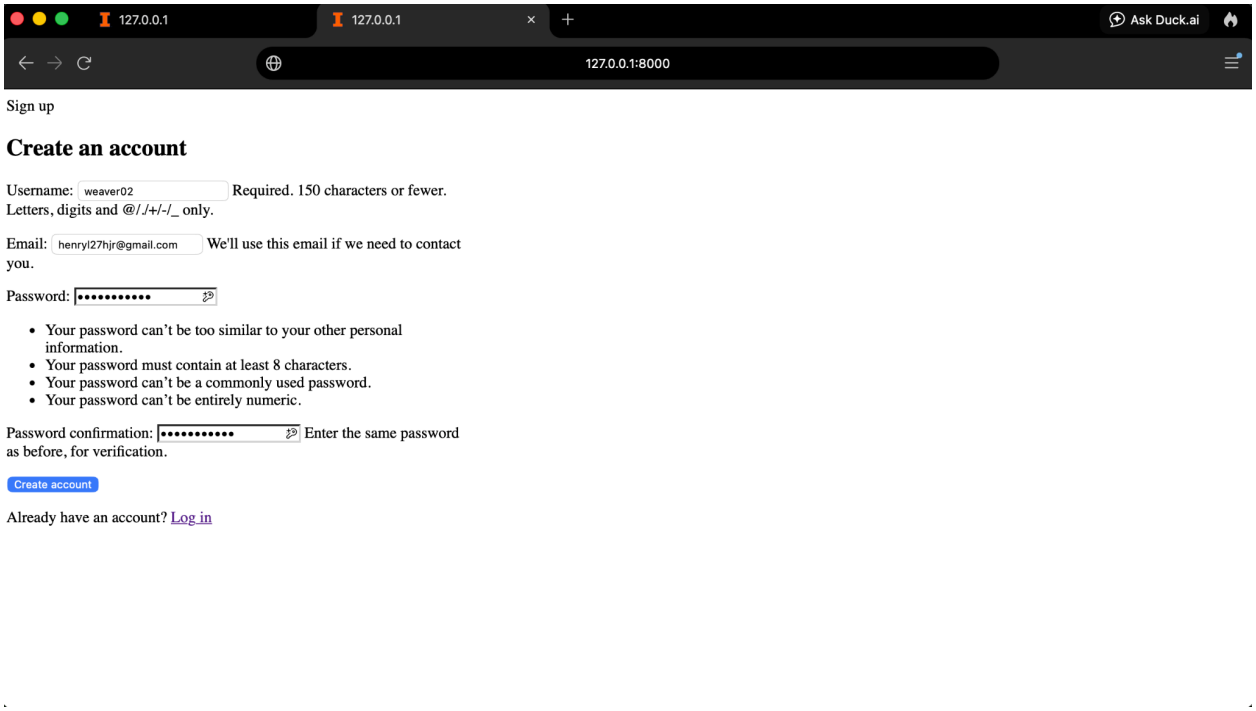
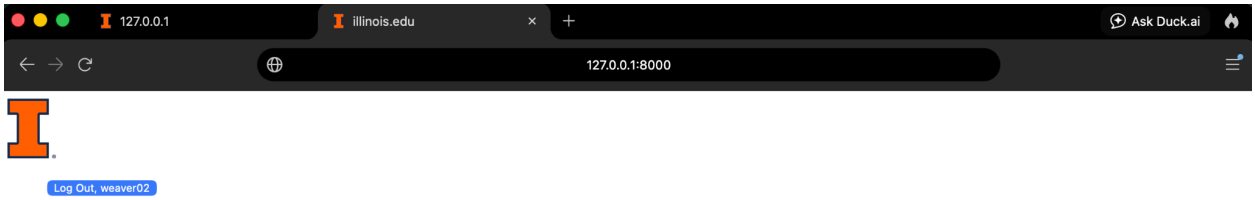
P1-A5
Weavers

Hosted: https://henry2jam.pythonanywhere.com/

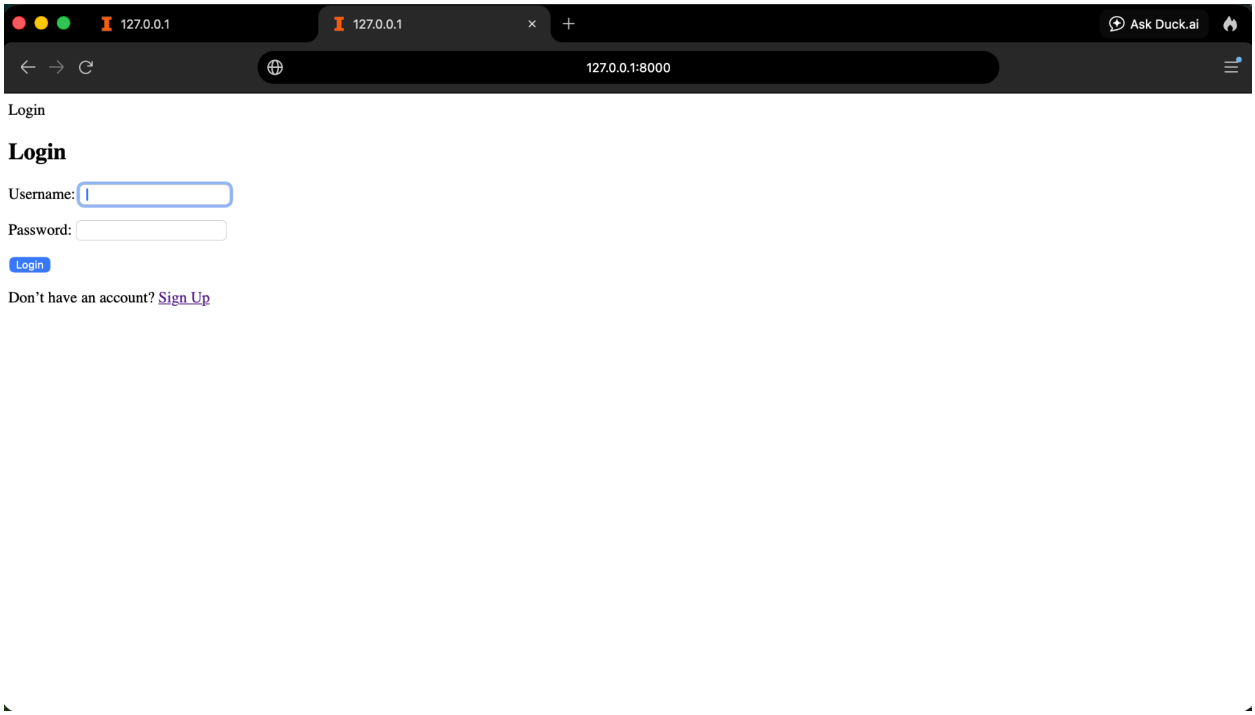
Part 1: Internal Django Authentication



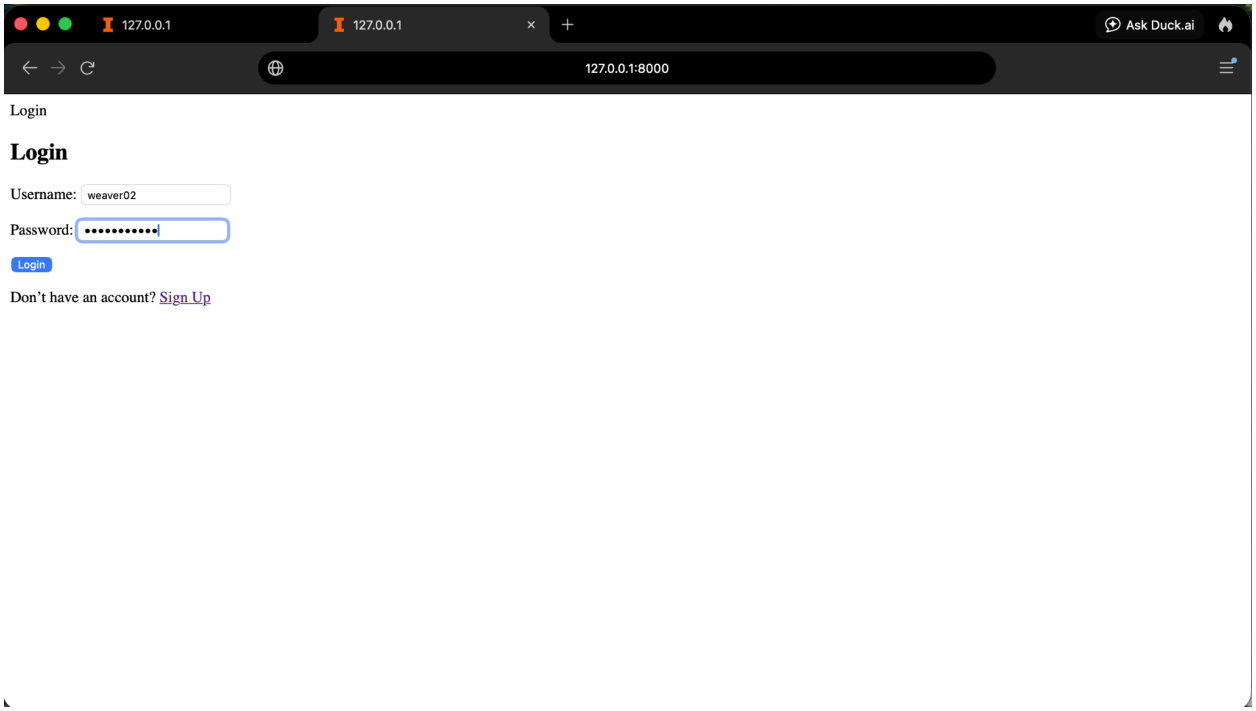
Sign up



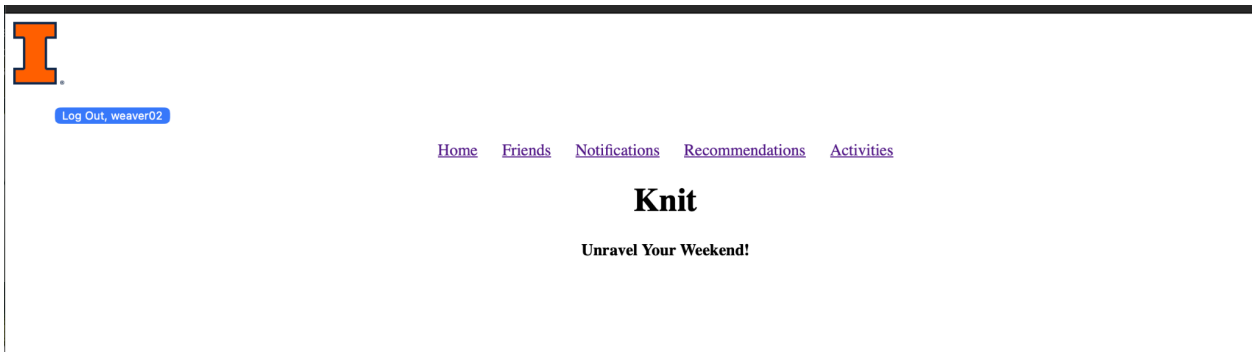
Successful account creation



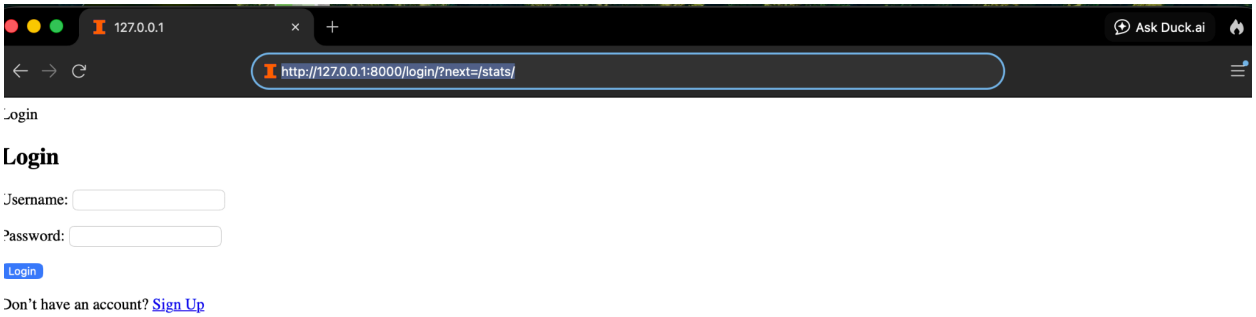
Clicked the “Log out, weaver02” button



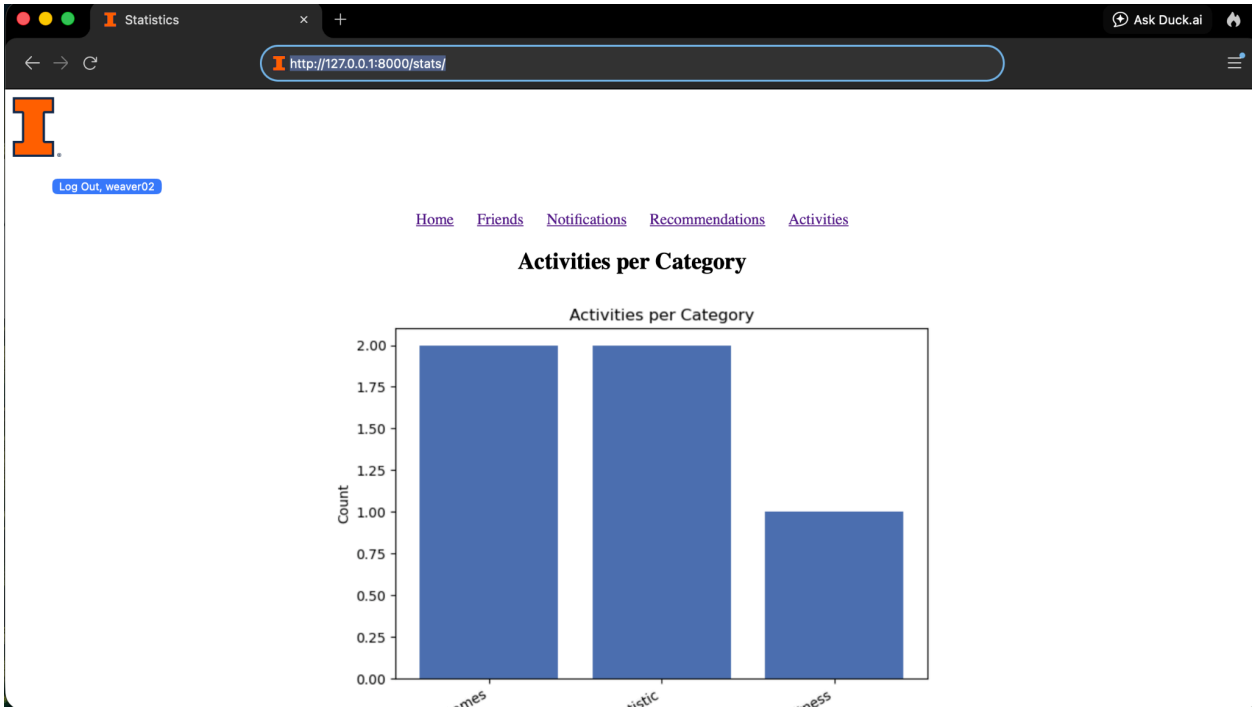
Login attempt



Success!

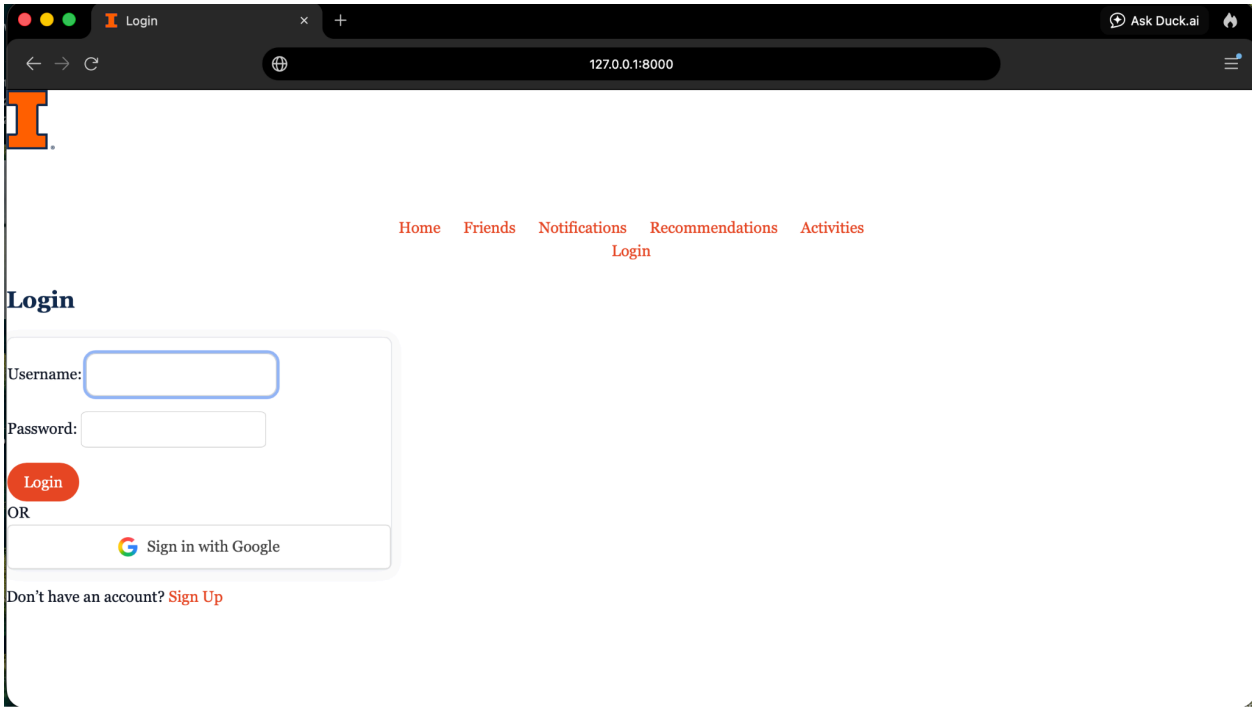


Applied protection to pages before login

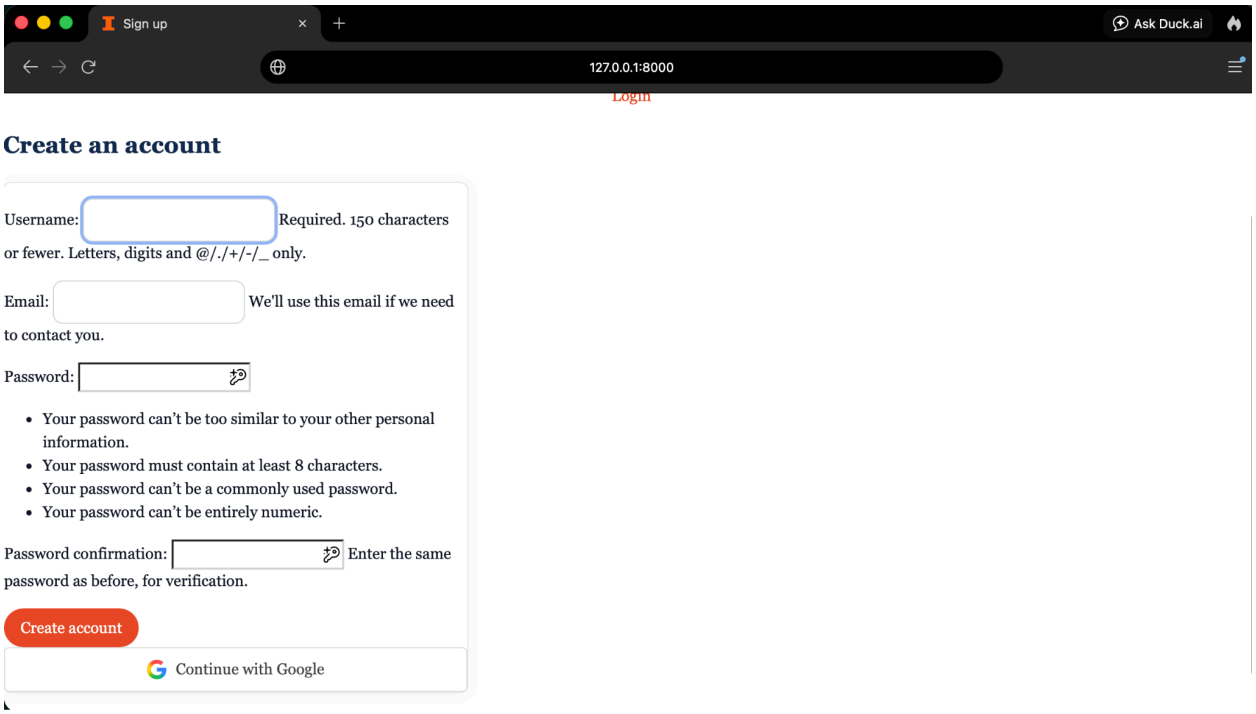


Accessible after login

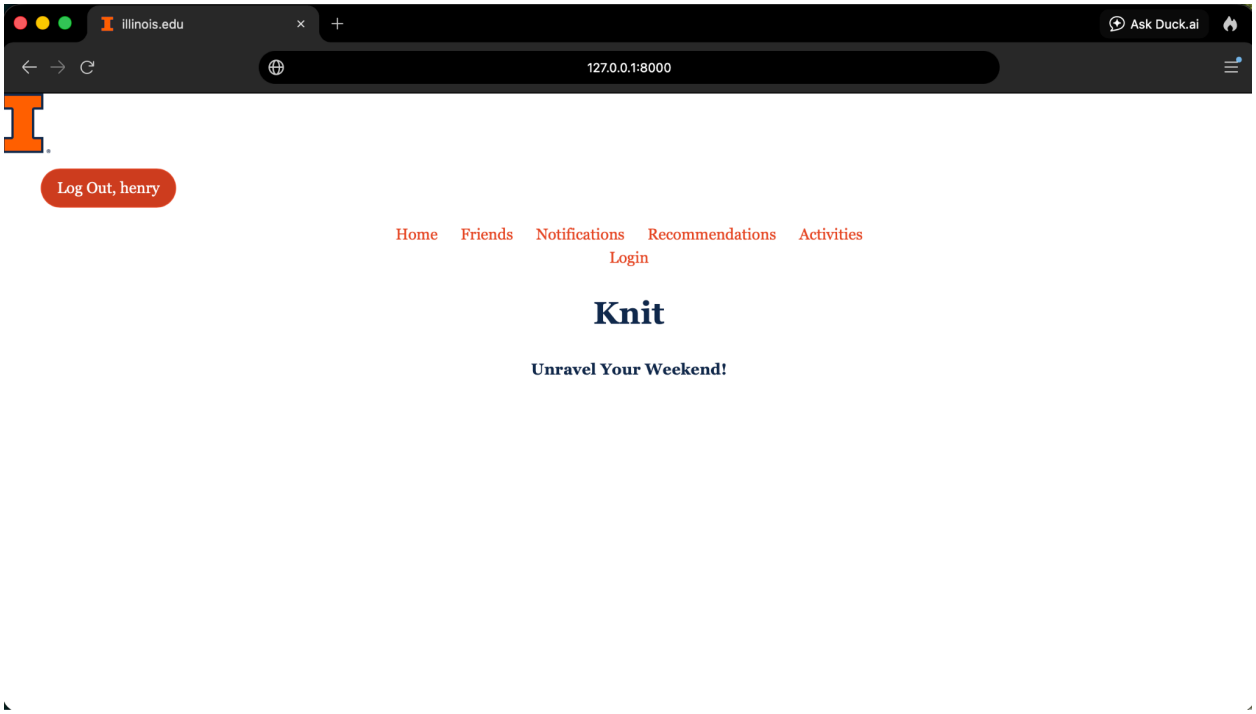
Part 2: Google OAuth



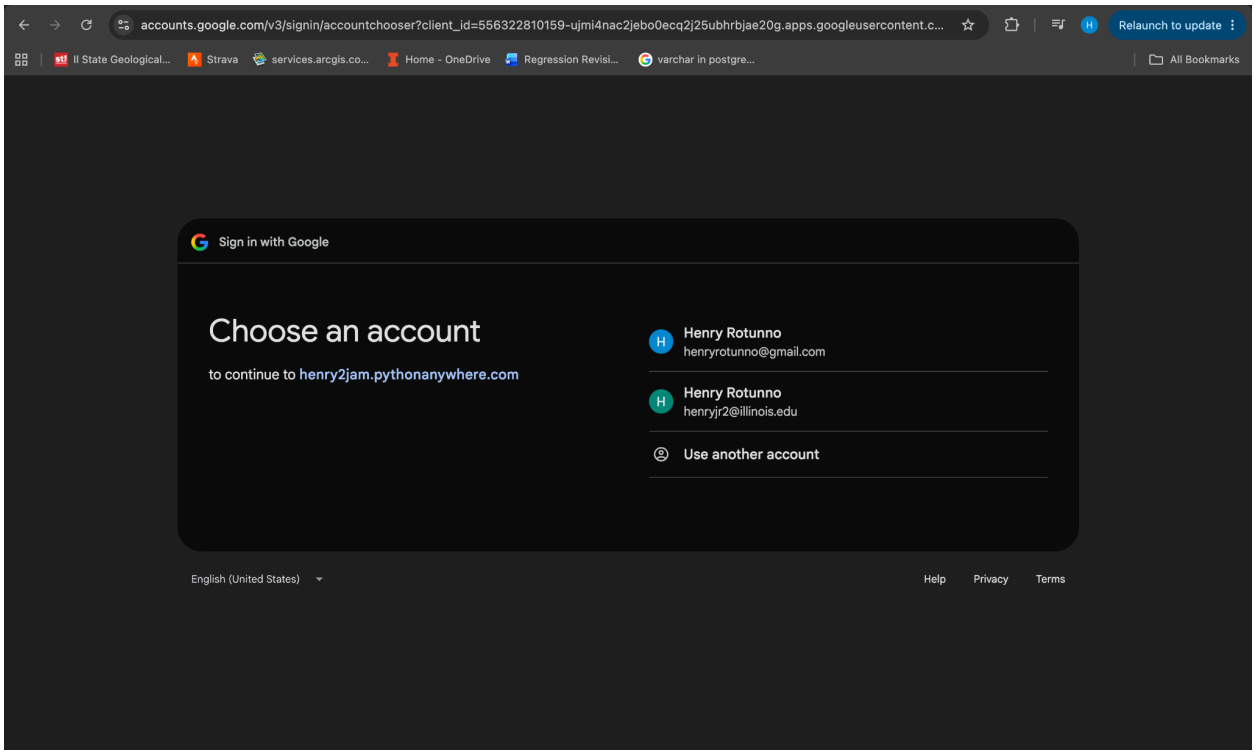
Implemented google button



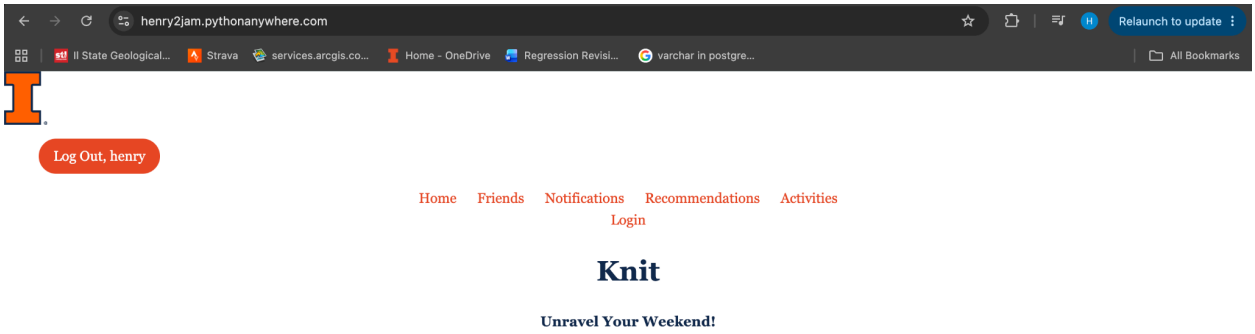
Signup with google



Works locally



Python anywhere OAuth working



Successful login on pythonanywhere

Note: We learned that pythonanywhere only allows calls to certain external apis. Unfortunately, the api in the recommendations page is not one of them. Attempting to use this feature results in a 403 which is handled in our try-except statement (recommendation view (views.py)).

Assignment Part 3

For whoever takes 3.4, here are the endpoint details you'll need:

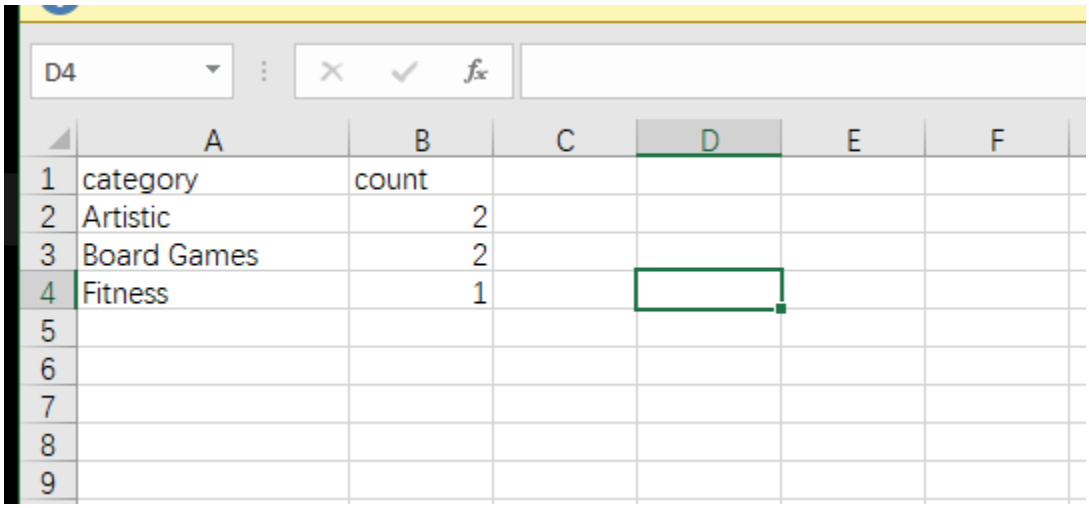
3.4 Alternative Uses of the Public API

Use Case 1: Python Summary



A Python script accesses the public activity-category API and retrieves the JSON data. The script calculates the total number of activities and identifies a category with the highest activity count. This demonstrates how an external application can use the API to generate a simple summary without accessing the Knit website directly.

Use Case 2: CSV Export for Excel



	A	B	C	D	E	F
1	category	count				
2	Artistic	2				
3	Board Games	2				
4	Fitness	1				
5						
6						
7						
8						
9						

A Python script retrieves the public API data and exports it into a CSV file. The CSV file can then be opened in Excel or imported into Google Sheets for further viewing and analysis. This demonstrates how the API can provide data for spreadsheet-based workflows.

Use Case 3: Statistical Analysis



```
(.venv) PS C:\Users\11200\PycharmProjects\8_Knit> python api_use_cases/use_case_3_statistics.py
Public API: https://kaiyunh2.pythonanywhere.com/api/chart/activity-categories/
API data: [{'category': 'Artistic', 'count': 2}, {'category': 'Board Games', 'count': 2}, {'category': 'Fitness', 'count': 1}]

Statistical Analysis
Total activities: 5
Artistic: 2 activities (40.0%)
Board Games: 2 activities (40.0%)
Fitness: 1 activities (20.0%)
(.venv) PS C:\Users\11200\PycharmProjects\8_Knit>
```

A Python script accesses the same public API and calculates the percentage of total activities represented by each category. Artistic and Board Games each represent 40% of the activities, while Fitness represents 20%. This demonstrates how the public API can be reused for basic statistical analysis.

Live Public Endpoint:
https://kaiyunh2.pythonanywhere.com/api/chart/activity-categories/
Method: GET (No authentication required)
Response Format:
JSON

```
[
  {"category": "Artistic", "count": 2},
  {"category": "Board Games", "count": 2},
  {"category": "Fitness", "count": 1}
]
```